

ZAKI HAIDER

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PROFESSIONAL SUMMARY

Final-year Software Engineering student at FAST-NUCES (completing 6th Semester) with hands-on expertise in AI, Machine Learning, Computer Vision, and Generative AI. Proven through national competition wins, research internships, and end-to-end AI projects. Actively seeking AI Engineering internships and roles where I can build intelligent, real-world systems that create meaningful impact.

WORK EXPERIENCE

Knowledge Discovery and Data Science Lab, FAST-NUCES

Jun 2025 – Aug 2025

Summer Intern — AI, ML & Computer Vision

- Built an end-to-end deep learning pipeline for text-to-image generation, achieving **91%+ accuracy** on a curated dataset of **100,000+ samples** without relying on any external APIs.
- Trained custom generative models from scratch, optimizing architecture and hyperparameters for efficiency and output quality.
- Gained deep hands-on experience in diffusion-based models, CNNs, and Vision Transformers (ViTs) for image synthesis tasks.

openHPI — Hasso Plattner Institute

May 2025

Practical Computer Vision with PyTorch — Trainee

- Implemented CNNs, Vision Transformers (ViTs), and diffusion-based models for object detection and image segmentation using PyTorch.
- Completed structured Computer Vision coursework covering modern deep learning architectures and practical model deployment techniques.

EDUCATION

FAST-NUCES, Islamabad, Pakistan

Aug 2023 – Jun 2027 (Expected)

Bachelor of Software Engineering | CGPA: 3.1+

Relevant Coursework: AI & Machine Learning, Neural Networks, Computer Vision, Data Mining, Database Management Systems, Software Quality Assurance, Operating Systems.

PROJECTS

Automatic University Timetable Generator — Python, Genetic Algorithms, Optimization

- Designed and implemented an intelligent scheduling system using Genetic Algorithms to automatically resolve conflicts across rooms, faculty, and time slots.
- Engineered fitness functions and crossover/mutation strategies to optimize schedule quality across multiple hard and soft constraints simultaneously.

CDA Transit Route Analyzer — Python, Process Mining, Event Logs, Map Integration, Chatbot

- Built a full transit analysis system integrating real CDA bus route maps; users input source and destination and receive optimized route suggestions with timings and departures.
- Applied Process Mining concepts (event log analysis) to model and analyse real-world transit patterns and passenger flows.
- Integrated an AI chatbot interface for natural-language route queries and travel assistance.

Gesture-Controlled Drawing Application — Python, OpenCV, MediaPipe, NumPy

- Developed a real-time hand-tracking application using MediaPipe and OpenCV for webcam-based drawing, shape recognition, and intuitive gesture-driven UI controls.

Career Connect — Job Fair Management System — SQL Server, Desktop Application

- Designed a 3NF-normalized SQL Server database and desktop application featuring multi-role access, real-time check-ins, and an admin dashboard for placement and interview analytics.

ACHIEVEMENTS

- **1st Position — App Development, SOFTEC (FAST University Lahore):** Secured first place at SOFTEC, a prestigious national-level 24-hour hackathon held at FAST Lahore, competing against top university teams from across Pakistan.
- **Winner — University-Wide AI Hackathon, SES Society:** Outperformed multi-university teams by developing a functional AI solution within a limited timeframe, demonstrating rapid prototyping and collaborative problem-solving skills.
- **4× Teaching Assistant — FAST-NUCES Islamabad:** Selected for four consecutive terms to assist faculty in core computing courses, mentoring junior students in programming logic, debugging, and software engineering principles.
- **Volunteer — Indus AI Week, Ministry of IT Pakistan:** Contributed as a technical volunteer at the national AI awareness event organised by the Ministry of IT, February 2026.

TECHNICAL SKILLS

AI & Machine Learning: Supervised & Unsupervised Learning, Deep Learning, Neural Networks, NLP, Generative AI, Diffusion Models, CNNs, Vision Transformers (ViTs), RAG Pipelines, Similarity Search, Genetic Algorithms, Process Mining

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, LangChain, HuggingFace, OpenCV, MediaPipe, Pandas, NumPy, Streamlit, Flask

Databases: MongoDB, MySQL, SQLite, PostgreSQL, ChromaDB

Programming: Python, C++, Java, JavaScript, SQL, HTML, CSS

No-Code / Automation: n8n (workflow automation), NocoDB (no-code database)

Tools: Git/GitHub, VS Code, Cypress, Jupyter Notebook

CERTIFICATIONS

- **Practical Computer Vision with PyTorch** | Hasso Plattner Institute (openHPI) | May 2025
- **Fundamentals of Machine Learning and Artificial Intelligence** | AWS Training & Certifications | Jun 2025
- **Process Mining: Data Science in Action** | Coursera — Eindhoven University of Technology | 2025
- **1st Place — App Development Hackathon Certificate** | SOFTEC, FAST University Lahore | 2025

LANGUAGES

Urdu: Native **English:** Intermediate Proficiency

References available upon request.